

1)

a)

In 1D: Charge density gives how much charge there is over a line per unit length, while the line current density gives how much charge is passing over a given time per unit length (current per unit length).

In 2D: Charge density gives how much charge there is over a surface per unit area, while the surface current density describes how much of the current running parallel to the flow is going through a "ribbon" on said surface, giving a current per unit width.

In 3D: Charge density gives how much charge there is in a bulk per unit volume, while the volume current density describes how much of the current running parallel to the flow is going through a "tube" in said bulk, giving a current per unit area.

b)

For a wire: Mobile charges are described with λ meaning the density of charges moving across the wire, correlating to the current.

For a thin conducting sheet: Mobile charges are described with σ meaning the density of charges moving across the sheet, correlating to the current.

For a bulk conducting material: Mobile charges are described with ρ meaning the density of charges moving through the material, correlating to the current.

c)

For a wire: $I = \lambda v$ is the current for a mobile charge.

For a thin conducting sheet: $K = \sigma v$ is the current for a mobile charge.

For a bulk conducting material: $J = \rho v$ is the current for a mobile charge.

d)

For a wire:

$$B(r) = \frac{\mu_0}{4\pi} I \int \frac{dl \times \hat{r}}{r^2}$$

For thin conducting sheet:

$$B(r) = \frac{\mu_0}{4\pi} \int \frac{K(r') \times \hat{r}}{r^2} d\alpha$$

For a bulk conducting material:

$$B(r) = \frac{\mu_0}{4\pi} \int \frac{J(r') \times \hat{r}}{r^2} d\tau$$

e)

For 1D, the mobile and constant charge are the same expression: $I = \lambda v$, while

	2D	3D
Constant	$K = \frac{dI}{dl}$	$J = \frac{dI}{d\alpha_{\perp}}$
Mobile	$K = \sigma v$	$J = \rho v$

differ between the constant and mobile charge flow expressions.

f)

λ , σ and ρ can appear in both electrostatics and magnetostatics as forms of representation of charges in 1D, 2D and 3D respectively. However they differ in terms of how the charges are over time, we assume constant charges densities in electrostatics while mobile in magnetostatics.

2)

a)

If more charge flows out through the surface of a volume, then the charge inside the volume will decrease.

b)

The flux through the surface S is given as

$$\oint_S J \cdot d\alpha = -\frac{\partial}{\partial t} \int_V \rho d\tau$$

and must be equal to the negative change of charge density over time. Using the divergence theorem we get:

$$\int_V (\nabla \cdot J) d\tau = -\frac{\partial}{\partial t} \int_V \rho d\tau$$

Which should be the same for any volume so we can remove the integrals and get:

$$\nabla \cdot J = -\frac{\partial \rho}{\partial t}$$

c)

The divergence of the current through the volume is equal to the negative change of charge density through time. This means that the rate of the current's flux through the volume is the same as the change of charge density through time.

d)

In magnetostatics, we work with steady currents that give constant magnetic fields (aka magnetostatics.) This means that the change of charge density is zero.

$$\implies \nabla \cdot J = -\frac{\partial \rho}{\partial t} = 0$$

e)

If the divergence of the volumetric current is not 0 in some region, the magnetic field created would not be constant. This could be for example an inductor given AC current, making the current vary from time and frequency.

3)

a)

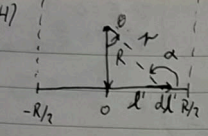
$$J = \frac{dI}{d\alpha_{\perp}} = \frac{I}{ab}$$

b)

$$I = \int_S J(x) \cdot d\alpha = \int_0^b \int_0^a kx \, dx dy = \frac{k}{2} a^2 b$$

4)

4) Looking at just one side of the square loop of wire.



From the right hand rule we get that $(d\mathbf{l} \times \mathbf{r})$ points out of the page, and from the definition of cross product ($\mathbf{A} \times \mathbf{B} = A \cdot B \cdot \sin \alpha$) we get the magnitude is: $dl \cdot 1 \cdot \sin \alpha = dl \cos \theta$ because θ and α are complementary angles.

and: $\sin \theta = \frac{l'}{r}$ & $\cos \theta = \frac{R}{r}$ Dividing these:

$$\Rightarrow \tan \theta = \frac{l'}{R} \Rightarrow l' = R \tan \theta \quad \text{Derivating this}$$

$$\Rightarrow dl' = \frac{R}{\cos^2 \theta} d\theta \quad \text{and we can turn around the cos:} \quad \left| \tan \theta' = \frac{1}{\cos^2 \theta} \right|$$

$$\cos \theta = \frac{R}{r} \Rightarrow \frac{\cos \theta}{R} = \frac{1}{r^2} \Rightarrow \frac{1}{r^2} = \frac{\cos^3 \theta}{R^2}$$

Now we can calculate: $B = \frac{\mu_0}{4\pi} I \int \frac{(d\mathbf{l} \times \mathbf{r})}{r^2}$

$$\Rightarrow B = \frac{\mu_0}{4\pi} I \int_{\theta_1}^{\theta_2} \left(\frac{\cos^2 \theta}{R^2} \right) \left(\frac{R}{\cos^2 \theta} \right) \cos \theta d\theta \quad \left| \begin{array}{l} \text{Turn integral} \\ \text{to angular} \end{array} \right|$$

$$\Rightarrow B = \frac{\mu_0}{4\pi} I \int_{\theta_1}^{\theta_2} \cos \theta d\theta = \frac{\mu_0}{4\pi} I (\sin \theta_2 - \sin \theta_1)$$

For the segment $[0, R/2]$ we got that $l' = R/2 = R \tan \theta_2 \Rightarrow \tan \theta_2 = \frac{1}{2} \Rightarrow \theta_2 = 0,4636$ and that $\theta_1 = 0$ so that the magnetic field becomes

$$B = \frac{\mu_0}{4\pi} I (\sin(0,4636) - \sin 0) = \frac{\mu_0}{4\pi} I \cdot 0,45$$

All the segments will contribute the same way in the point R, so we can multiply it by 8 (4 half sides) and get

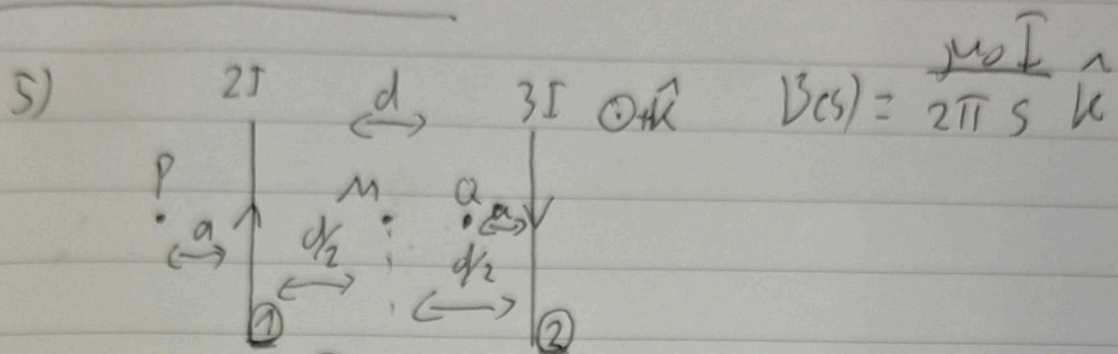
$$B = \frac{2}{\pi} \mu_0 I \cdot 0,45$$

5)

Using the formula for magnetic force from last task for an infinite long wire, the angles θ_1 and θ_2 become $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ respectively, leading to:

$$B(s) = \frac{\mu_0 I}{2\pi s} \hat{k}$$

By using the right hand rule to determine the direction of the two magnetic fields and summing them we can calculate the magnetic fields and then use Lorentz force law for line integral over a unit length L to find the force per unit length:



$$a) B_p = \left(\frac{\mu_0 2I}{2\pi a} - \frac{\mu_0 3I}{2\pi (a+d)} \right) \hat{k}$$

$$b) B_Q = \left(\frac{\mu_0 2I}{2\pi (d-a)} + \frac{\mu_0 3I}{2\pi a} \right) (-\hat{k})$$

$$c) B_M = \left(\frac{\mu_0 2I}{2\pi \cdot d/2} + \frac{\mu_0 3I}{2\pi d/2} \right) \hat{k} = \frac{\mu_0 5I}{2\pi d} \hat{k}$$

Field on ② is

$$d) B = \frac{\mu_0 2I}{2\pi d} \Rightarrow F = 3I \left(\frac{\mu_0 2I}{2\pi d} \right) \int_0^L dl \Rightarrow \underline{\underline{\frac{F}{L} = \frac{\mu_0}{2\pi} \cdot \frac{6I^2}{d}}}$$

Since the currents are in the opposite direction, they will repel each other.